A FUNCTION SEPARATING THE COMPONENTS OF THE SPECIAL FIBRE AT CHOSEN POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the choice of the points avoiding the other components, the local elements cutting out the components, and the multiplication of the numerators of the local expressions.

Editorial note. The label lemma-nice-meromorphic-function names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-nice-meromorphic-function –start 1622 –end 1641 –local-id 28120 returned RESULT=OK (L28120 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The fraction fields of the base and of the local rings of the source agree. The observations following the vanishing situation, describing the source as a modification of the spectrum of the base and the relation between the function fields. Allows writing elements of the maximal ideal of a local ring of the source as fractions of elements of the base.
- Choice of a closed point of one component outside the others. The observation that the components of the special fibre are pairwise distinct closed curves, so that a closed point of one of them can be chosen avoiding the others. Produces the points at which the factorization of the constructed function is required.

Source reference. Original reference 07UY: sources/stacks-resolve/pinned-original-source.tex, source lines 1584–1584. Prerequisite pointer invoked at proof line(s) 1624; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 07UY there exists a nonzero $f \in \mathfrak{m}$ such that for every $i = 1, \ldots, r$ there exist*

(1) *a closed point $x_i \in C_i$ with $x_i \notin C_j$ for $j \neq i$,*

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(2) *a factorization $f = g_i f_i$ of f in $\mathcal{O}_{X,x_i}$ such that $g_i \in \mathfrak{m}_{x_i}$ maps to a nonzero element of $\mathcal{O}_{C_i,x_i}$.*

Proof. We will use the observations made following Situation 07UY without further mention. Pick a closed point $x_i \in C_i$ which is not in C_j for $j \neq i$. Pick $g_i \in \mathfrak{m}_{x_i}$ which maps to a nonzero element of $\mathcal{O}_{C_i,x_i}$. Since the fraction field of A is the fraction field of $\mathcal{O}_{X_i,x_i}$ we can write $g_i = a_i/b_i$ for some $a_i, b_i \in A$. Take $f = \prod a_i$. □